

AYON, Russian Federation

WMO: 250420

Lat: 69.935N

Lon: 167.989E

Elev: 45

StdP: 14.67

Time Zone: 12.00 (E12)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -38.7	(c) -35.6	(d) -43.4	(e) 0.4	(f) -38.3	(g) -40.5	(h) 0.5	(i) -35.3	(j) 32.1	(k) -4.7	(l) 29.6	(m) 6.6	(n) 6.3	(o) 170	(p) 1.073

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	67.1	56.6	62.7	54.7	57.9	51.8	57.8	65.1	54.9	62.3	51.9	57.8	14.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
53.3	60.5	63.7	50.6	54.7	59.3	47.6	48.8	55.7	25.0	65.0	23.2	62.5	21.4	57.9	64.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
32.1	27.3	23.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			-40.6	60.5	3.2	4.6	-42.8	63.8	-44.7	66.6	-46.5	69.2	-48.8	72.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.3	-16.0	-14.7	-12.3	4.8	22.9	38.3	43.2	41.1	34.7	21.4	3.3	-9.1
	DBStd	24.47	11.51	12.72	11.82	10.87	8.11	8.11	8.41	8.00	6.41	10.35	12.59	13.71
	HDD50	13492	2047	1810	1932	1357	839	367	254	302	463	887	1400	1834
	HDD65	18880	2512	2230	2397	1807	1304	800	677	743	909	1352	1850	2299
	CDD50	90	0	0	0	0	0	18	43	25	4	0	0	0
	CDD65	2	0	0	0	0	0	0	1	1	0	0	0	0
	CDH74	21	0	0	0	0	0	0	16	5	0	0	0	0
	CDH80	2	0	0	0	0	0	0	2	0	0	0	0	0
	WSAvg	11.1	9.9	10.2	11.1	11.0	10.4	10.7	10.7	11.7	12.2	11.7	13.0	10.9
	PrecAvg	4.4	0.2	0.3	0.2	0.3	0.2	0.3	0.5	0.6	0.6	0.6	0.4	0.3
(15) Precipitation	PrecMax	6.5	0.7	0.6	0.4	0.7	1.2	0.7	1.5	1.5	0.9	1.6	1.0	1.0
	PrecMin	2.2	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.2	0.2	0.1	0.1	0.0
	PrecStd	0.9	0.2	0.2	0.1	0.2	0.2	0.2	0.3	0.4	0.2	0.3	0.2	0.2
	PrecStd	0.9	0.2	0.2	0.1	0.2	0.2	0.2	0.3	0.4	0.2	0.3	0.2	0.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	24.8	24.9	26.7	34.2	46.4	69.6	75.1	72.9	62.4	42.8	30.8	28.2	27.4
	0.4% MCWB	23.6	23.5	25.8	32.5	43.4	55.6	62.3	61.8	53.2	40.9	29.9	27.3	27.3
	2% DB	17.0	16.4	16.6	28.7	37.7	64.7	68.4	64.6	51.1	37.7	28.4	20.9	20.9
	2% MCWB	16.2	15.7	15.8	27.3	35.9	53.6	58.3	56.0	46.6	37.1	27.6	20.2	20.2
	5% DB	4.1	7.8	10.6	24.4	35.5	59.6	63.7	59.7	46.5	35.5	24.6	16.2	16.2
	5% MCWB	3.5	7.1	10.1	23.4	33.9	51.4	55.9	53.6	43.6	34.9	23.6	15.6	15.6
	10% DB	-2.4	3.0	5.1	20.7	33.7	53.0	58.3	54.5	42.7	33.2	21.0	10.1	10.1
	10% MCWB	-2.9	2.6	4.7	19.8	32.5	47.4	53.0	50.0	40.7	32.4	20.2	9.5	9.5
	0.4% WB	23.3	23.5	26.1	32.4	43.8	56.6	62.6	63.0	53.1	41.2	30.2	27.4	27.4
	0.4% MCDB	24.7	24.4	27.1	34.1	46.6	67.4	74.9	72.6	61.3	42.6	30.5	28.5	28.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	16.3	15.9	15.8	27.4	35.8	54.0	58.8	57.2	47.4	37.3	27.5	20.3	20.3
	2% MCDB	17.0	16.8	16.4	28.8	37.3	63.9	66.6	63.0	50.8	37.8	28.4	20.8	20.8
	5% WB	3.4	7.2	10.1	23.4	34.1	51.0	56.1	53.4	43.6	35.0	23.6	15.6	15.6
	5% MCDB	3.9	7.8	10.5	24.3	35.2	59.3	63.1	59.2	46.0	35.4	24.5	16.2	16.2
	10% WB	-2.8	2.6	4.6	19.8	32.6	47.2	53.1	50.3	40.8	32.6	20.1	9.7	9.7
	10% MCDB	-2.4	2.9	5.1	20.6	33.6	53.1	58.4	54.8	42.4	33.2	21.1	10.3	10.3
(35) Mean Daily Temperature Range	MDBR	9.5	9.8	10.9	12.4	9.6	11.3	12.6	10.5	7.4	7.0	9.7	9.3	9.3
	5% DB	18.4	15.8	14.4	13.9	10.1	24.5	23.3	21.8	15.1	6.1	9.8	13.4	13.4
	5% WB	17.9	15.6	14.2	13.0	8.9	15.5	14.3	14.1	10.9	5.7	9.9	13.1	13.1
	MCDBR	18.5	15.8	14.6	13.7	9.5	23.9	22.2	20.7	14.1	5.4	10.0	13.3	13.3
	MCWBR	18.0	15.6	14.5	12.8	8.6	15.6	14.1	14.0	10.3	5.0	10.1	13.0	13.0
	MCWBR	18.0	15.6	14.5	12.8	8.6	15.6	14.1	14.0	10.3	5.0	10.1	13.0	13.0
(40) Clear-Sky Solar Irradiance	taub	N/A	0.207	0.283	0.324	0.349	0.344	0.377	0.376	0.312	0.248	N/A	N/A	N/A
	taud	N/A	2.009	2.016	1.962	2.011	2.226	2.293	2.325	2.466	2.135	N/A	N/A	N/A
	Ebn at Noon	0	169	221	249	259	267	252	236	225	145	0	0	0
	Edh at Noon	0	13	27	40	43	36	33	29	18	11	0	0	0
(44) All-Sky Solar Radiation	RadAvg	0	108	633	1383	1991	2056	1607	1034	521	154	0	0	0
	RadStd	0	18	37	66	90	176	132	91	35	21	0	0	0

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page